Electric Field

Coulomb's Force Between Two Point Charges

Consider two point charges Q_1 and Q_2 located at position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$. Define the displacement vector from Q_1 to Q_2 as

$$\mathbf{R}_{12} = \mathbf{r}_2 - \mathbf{r}_1, \qquad R_{12} = \|\mathbf{R}_{12}\|, \qquad \hat{\mathbf{a}}_{12} = \frac{\mathbf{R}_{12}}{R_{12}}.$$

Coulomb's law gives the force on Q_2 due to Q_1 as

$$\mathbf{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{Q_1 Q_2}{R_{12}^2} \hat{\mathbf{a}}_{12}.$$

Proportionality with the test charge. Since Q_2 appears linearly,

$$\mathbf{F}_{12} \propto Q_2, \qquad \text{so if } Q_2 \to 4Q_2, \text{ then } \mathbf{F}'_{12} = 4\,\mathbf{F}_{12}.$$

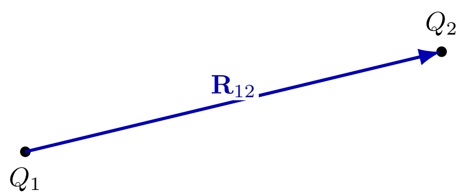


Figure 1: Two charges separated by $\mathbf{R}_{12}$ (from Q_1 to Q_2).

A Useful Definition: Electric Field

A convenient and physically meaningful definition is:

Force per coulomb of Q_2 $\Rightarrow$ **Electric field.**

$$\boxed{\mathbf{E}(\mathbf{r}_2) = \frac{1}{4\pi\varepsilon_0} \frac{Q_1}{R_{12}^2} \hat{\mathbf{a}}_{12} \quad (\text{units: N/C})}$$

Equivalently,

$$\mathbf{E} = \frac{\mathbf{F}}{Q_2}.$$

After introducing the electrostatic potential, we often express field units as

$$\text{V/m} \quad (\text{since } 1 \text{ N/C} = 1 \text{ V/m}).$$

Why Introduce a New Vector E If We Already Know $\mathbf{F}_{12}$?

If we already know the force vector $\mathbf{F}_{12}$, why do we need a new vector $\mathbf{E}$ (electric field)?

Key idea: It has something to do with how we interpret Coulomb's law.

Field ("stress") interpretation:

- The presence of the charge Q_1 *polarizes the material* (or, more generally, modifies the surrounding medium/space).
- In the presence of charge, the material can be thought of as getting *stressed*.
- Any time we place another charge Q_2 in this stressed material, it receives a "kick". This kick is what we call *force*.

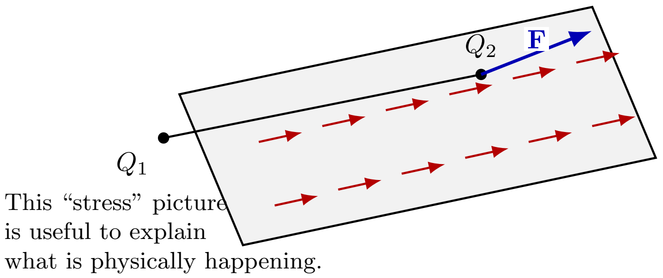


Figure 2: Heuristic picture: Q_1 creates a "stressed" environment; inserting Q_2 produces/experiences a force $\mathbf{F}$. The electric field $\mathbf{E}$ captures the environment independently of the chosen test charge.

From "force per Coulomb" to a field defined everywhere

For two charges Q_1 at $\mathbf{r}_1$ and Q_2 at $\mathbf{r}_2$, define

$$\mathbf{R}_{12} = \mathbf{r}_2 - \mathbf{r}_1, \qquad R_{12} = \|\mathbf{R}_{12}\|, \qquad \hat{\mathbf{a}}_{12} = \frac{\mathbf{R}_{12}}{R_{12}}.$$

The electric field at the location of Q_2 due to Q_1 is

$$\mathbf{E}(\mathbf{r}_2) = \frac{1}{4\pi\varepsilon_0} \frac{Q_1}{R_{12}^2} \hat{\mathbf{a}}_{12}.$$

This is the electric field *experienced at the position of Q_2 due to Q_1*.

But here, we are not introducing the second charge.
We define the field due to Q_1 at *any arbitrary point in space.*

Field due to a single charge using an independent displacement vector

Let Q_1 be at the origin. Let $\mathbf{d}$ be the independent displacement vector from Q_1 to an arbitrary observation point:

$$\mathbf{d} = d_x\,\hat{\mathbf{a}}_x + d_y\,\hat{\mathbf{a}}_y + d_z\,\hat{\mathbf{a}}_z, \qquad d = \|\mathbf{d}\|, \qquad \hat{\mathbf{d}} = \frac{\mathbf{d}}{d}.$$

$$\boxed{\mathbf{E}(\mathbf{d}) = \frac{1}{4\pi\varepsilon_0}\,\frac{Q_1}{d^2}\,\hat{\mathbf{d}}}$$

Interpretation.

- It is similar to saying: at all points around Q_1, there exists a *potential force* waiting to happen.
- Now $\mathbf{E}(\mathbf{d})$ is a function of the independent variable $\mathbf{d}$ (which is itself a vector).
- Hence, the electric field is a **vector field**.
- This is essentially Coulomb's law, except it is written without explicitly placing a second charge—it applies to any point in space.

Superposition Principle

If there are N source charges $\{Q_i\}_{i=1}^{N}$ located at positions $\{\mathbf{r}_i\}_{i=1}^{N}$, and the observation point is at $\mathbf{r}$, define the displacement from Q_i to the observation point as

$$\mathbf{R}_i = \mathbf{r} - \mathbf{r}_i, \qquad R_i = \|\mathbf{R}_i\|, \qquad \hat{\mathbf{R}}_i = \frac{\mathbf{R}_i}{R_i}.$$

Then the total electric field is the vector sum of individual fields:

$$\boxed{\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N}\frac{Q_i}{R_i^2}\,\hat{\mathbf{R}}_i}$$

or equivalently,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N} Q_i\,\frac{\mathbf{R}_i}{R_i^3}.$$

If a test charge q is placed at $\mathbf{r}$, the force on it is

$$\boxed{\mathbf{F}(\mathbf{r}) = q\,\mathbf{E}(\mathbf{r})} \quad \Rightarrow \quad \mathbf{F}(\mathbf{r}) = \frac{q}{4\pi\varepsilon_0}\sum_{i=1}^{N}\frac{Q_i}{R_i^2}\,\hat{\mathbf{R}}_i.$$

or equivalently,

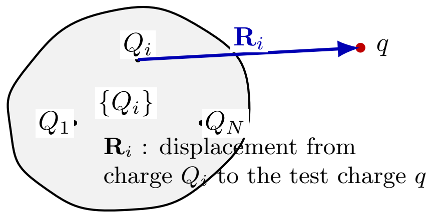


Figure 3: Superposition: the field (and hence force on a test charge q) is the vector sum of contributions from each Q_i.

$$\boxed{\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N} Q_i \frac{\mathbf{r}-\mathbf{r}_i'}{\|\mathbf{r}-\mathbf{r}_i'\|^3}} \qquad \text{(here } \mathbf{r} \text{ is the independent variable).}$$

Electric Field Due to a "Cloud" of Charges (Continuous Distribution)

In matter, charges are distributed in enormous numbers (typically $N \gg 10^{23}$). Instead of summing over individual charges, we model the distribution using a *charge density* (Ref. Fig.4).

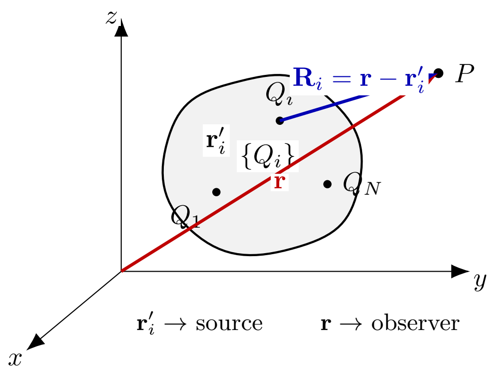


Figure 4: Electric field at $\mathbf{r}$ due to multiple discrete source charges Q_i located at $\mathbf{r}_i'$ (in the charged region V).

Coarse graining using a box of size Δ

Partition the charge region into small cubic boxes of side Δ (volume Δ^3). Let ΔQ_k be the net charge inside the k-th box, whose representative position is $\mathbf{r}_k'$ (Ref. Fig.5).

How ΔQ scales with Δ (and why the density can still be meaningful)

Assume the charge distribution is approximately uniform at the scale of interest, with (local) volume charge density ρ_0 (units $\mathrm{C/m^3}$). Then the charge contained in a cubic box of side Δ scales as

$$\Delta Q \;\approx\; \rho_0\,\Delta^3.$$

Hence, as Δ decreases, the *charge in the box* (and therefore the *number of charges*) decreases rapidly, even though the density (ideally) remains the same.

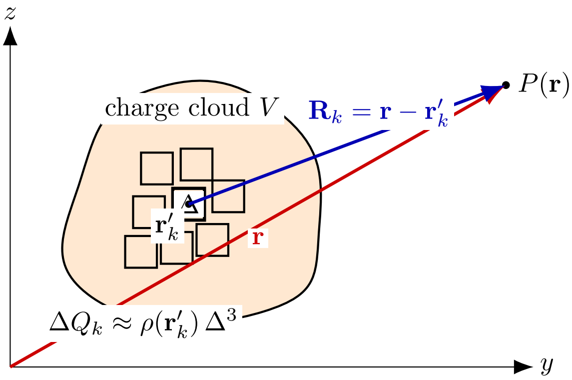


Figure 5: Coarse graining: partition the charged region V into boxes (cells) of side Δ. Each cell k is represented by its net charge ΔQ_k located at a representative point $\mathbf{r}'_k$, enabling a box-sum approximation for $\mathbf{E}(\mathbf{r})$.

Box size Δ	**Volume Δ^3 ($\mathrm{m^3}$)**	**Charge in box $\Delta Q \approx \rho_0\Delta^3$**	**Density estimate $\rho_{\text{est}} = \Delta Q/\Delta^3$**
$1\ \mathrm{cm} = 10^{-2}\ \mathrm{m}$	10^{-6}	$\rho_0 \times 10^{-6}$	ρ_0
$1\ \mathrm{mm} = 10^{-3}\ \mathrm{m}$	10^{-9}	$\rho_0 \times 10^{-9}$	ρ_0
$1\ \mu\mathrm{m} = 10^{-6}\ \mathrm{m}$	10^{-18}	$\rho_0 \times 10^{-18}$	ρ_0

Table 1: Decreasing Δ reduces the charge (and number of charges) inside a box as Δ^3, even if the underlying density is uniform.

Important practical point. While $\rho_{\text{est}} = \Delta Q/\Delta^3$ is *formally* constant for a perfectly uniform continuum, for very small Δ the box may contain only a few discrete charges (or sometimes none). Then ΔQ exhibits significant statistical fluctuations, and ρ_{est} becomes noisy. This is why, in practice, we choose Δ large enough to include many charges (so fluctuations average out), but small enough to resolve macroscopic spatial variation of $\rho(\mathbf{r}')$ (Ref. Fig6).

(A) Discrete-charge sum (exact, microscopic). If the charge cloud consists of N individual charges q_i located at $\mathbf{r}'_i$, then

$$\boxed{\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N} q_i\,\frac{\mathbf{r}-\mathbf{r}'_i}{\|\mathbf{r}-\mathbf{r}'_i\|^3}}$$

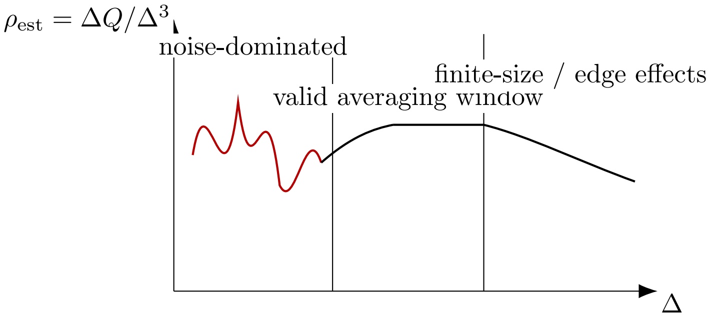


Figure 6: Qualitative behavior of the density estimate $\rho_{\rm est} = \Delta Q/\Delta^3$ versus averaging scale Δ. A nearly constant estimate is obtained only over an intermediate range of Δ.

This expression is exact, but in matter N is typically enormous ($N \gg 10^{23}$) and the microscopic positions $\mathbf{r}'_i$ fluctuate (thermal motion, collisions, etc.), so the sum is not practically useful for macroscopic prediction.

(B) Box (coarse-grained) sum (macroscopically meaningful). Partition the charge region into M boxes of side Δ. In box k, define the net charge

$$\boxed{\Delta Q_k \;\approx\; \sum_{i\in\text{box } k} q_i}$$

and choose a representative point (e.g., the box center) $\mathbf{r}'_k$. Then the field can be approximated by the *box sum*

$$\boxed{\mathbf{E}(\mathbf{r}) \approx \frac{1}{4\pi\varepsilon_0}\sum_{k=1}^{M} \Delta Q_k \,\frac{\mathbf{r}-\mathbf{r}'_k}{\|\mathbf{r}-\mathbf{r}'_k\|^3}}$$

Why the box sum is more relevant. Each ΔQ_k averages over many microscopic charges inside a box (so fluctuations reduce), and the number of terms M is vastly smaller than N. Therefore the box sum captures the *macroscopic* field that is stable and measurable, whereas the microscopic sum depends on inaccessible and rapidly changing microscopic details.

Continuum limit. If Δ is chosen in a valid averaging window (many charges per box, yet small compared to the spatial scale of variation), define $\rho(\mathbf{r}')$ by

$$\rho(\mathbf{r}') = \lim_{\Delta\to 0}\frac{\Delta Q}{\Delta^3}, \qquad \Delta Q_k \approx \rho(\mathbf{r}'_k)\,\Delta^3,$$

and the box sum tends to

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\iiint_V \rho(\mathbf{r}')\,\frac{\mathbf{r}-\mathbf{r}'}{\|\mathbf{r}-\mathbf{r}'\|^3}\,d\tau'.$$

Key Takeaways

- Microscopic description: $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N} q_i \frac{\mathbf{r}-\mathbf{r}'_i}{\|\mathbf{r}-\mathbf{r}'_i\|^3}$ (exact but impractical for $N \gg 10^{23}$).
- Coarse-grained (box) description: replace many charges in a box by ΔQ_k at $\mathbf{r}'_k$ and sum over boxes.
- Choose Δ in an intermediate averaging window: enough charges per box to reduce noise, yet small enough to resolve $\rho(\mathbf{r}')$ variation.
- Continuum limit: $\Delta Q_k \approx \rho(\mathbf{r}'_k)\Delta^3$ and the box sum tends to the volume integral.

Example 1: Electric field due to a finite line charge (linear charge density)

Problem. A straight line charge of uniform *linear charge density* λ (C/m) is placed along the z-axis from $z = -a$ to $z = +a$. Find the electric field at a point

$$P(\mathbf{r}) = (x, 0, 0),$$

i.e., at perpendicular distance x from the line.

Geometry and differential element

For a small element of length dz located at z, the charge contained in that element is

$$dq = \lambda\, dz.$$

The displacement from the source element at $(0, 0, z)$ to the observation point $P(x, 0, 0)$ is

$$\mathbf{R} = x\,\hat{\mathbf{x}} - z\,\hat{\mathbf{z}}, \qquad R = \|\mathbf{R}\| = \sqrt{x^2 + z^2}.$$

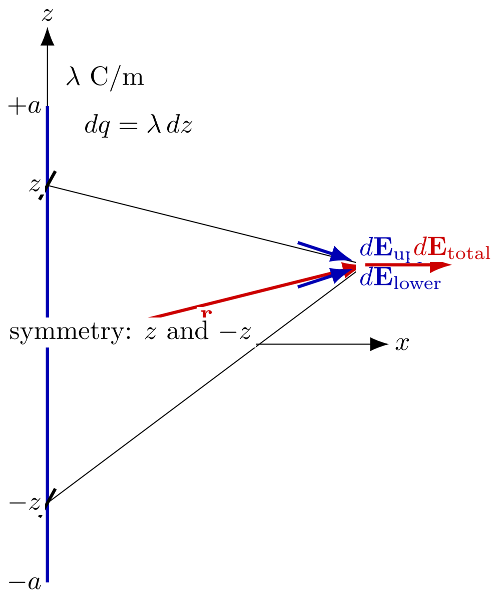
Field contribution and symmetry

Magnitude of the field due to one element $dq = \lambda dz$ at distance R is

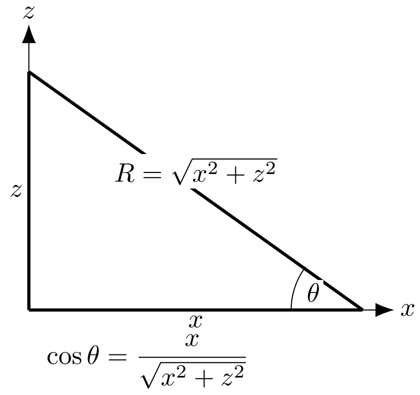
$$|d\mathbf{E}| = \frac{1}{4\pi\varepsilon_0}\frac{dq}{R^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dz}{x^2 + z^2}.$$

The two symmetric elements at $+z$ and $-z$ produce equal magnitudes:

$$|d\mathbf{E}_{\text{upper}}| = |d\mathbf{E}_{\text{lower}}| = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dz}{x^2 + z^2}.$$



Symmetric elements at z and $-z$: vertical components cancel; horizontal components add.



Geometry for one element at z: $R^2 = x^2 + z^2$ and $\cos\theta = x/R$.

Figure 7: Finite line charge along z-axis and observation point P at distance x.

Their z-components cancel, while the x-components add:

$$d\mathbf{E}_{\text{total}} = \Big(|d\mathbf{E}_{\text{upper}}|\cos\theta + |d\mathbf{E}_{\text{lower}}|\cos\theta\Big)\,\hat{\mathbf{x}} = 2\,|d\mathbf{E}|\,\cos\theta\,\hat{\mathbf{x}}.$$

Using

$$\cos\theta = \frac{x}{\sqrt{x^2 + z^2}},$$

we get

$$d\mathbf{E}_{\text{total}} = \frac{1}{4\pi\varepsilon_0}\,2\lambda\,dz\,\frac{x}{(x^2 + z^2)^{3/2}}\,\hat{\mathbf{x}}.$$

Integration over the line segment

Integrate from $z = 0$ to $z = a$ and multiply by 2 (already accounted for above by symmetry):

$$\mathbf{E}(x) = \hat{\mathbf{x}} \int_0^a \frac{1}{4\pi\varepsilon_0}\,2\lambda\,\frac{x}{(x^2 + z^2)^{3/2}}\,dz.$$

Using

$$\int \frac{dz}{(x^2 + z^2)^{3/2}} = \frac{z}{x^2\sqrt{x^2 + z^2}},$$

the result is

$$\boxed{\mathbf{E}(x) = \frac{1}{4\pi\varepsilon_0}\,\frac{2\lambda a}{x\sqrt{x^2+a^2}}\,\hat{\mathbf{x}}}$$

(pointing radially outward from the line for $\lambda > 0$).

Check (infinite line limit). As $a \to \infty$,

$$\mathbf{E}(x) \to \frac{\lambda}{2\pi\varepsilon_0 x}\,\hat{\mathbf{x}},$$

which matches the standard result for an infinite line charge.

Key Takeaways

- For a finite line segment from $-a$ to $+a$ on the z-axis, symmetry cancels the axial components; only radial components add.
- With linear charge density λ, at distance x from the line:

$$\mathbf{E}(x) = \frac{1}{4\pi\varepsilon_0}\frac{2\lambda a}{x\sqrt{x^2+a^2}}\,\hat{\mathbf{x}}.$$

- Infinite-line limit ($a \to \infty$): $\mathbf{E}(x) \to \frac{\lambda}{2\pi\varepsilon_0 x}\,\hat{\mathbf{x}}$.

Intuitive explanation : why $E \propto 1/r$ for a line charge

Let the (very long / infinite) line charge lie on the z-axis with uniform density λ. Let the observation point be at perpendicular distance r from the line (same as x in previous expression).

1) Symmetry: why vertical components cancel

Pick two equal charge elements at $+z$ and $-z$:

$$dq = \lambda\, dz.$$

At the observation point, their field magnitudes are equal, but their z-components are opposite. Hence,

$$dE_z(+z) + dE_z(-z) = 0, \qquad dE_r(+z) + dE_r(-z) = 2\,dE_r(z).$$

So the net field is purely radial (perpendicular to the line) and is obtained by adding the radial components.

2) Key idea: as r increases, a *longer portion of the line* contributes effectively

For an element at coordinate z, the distance to the observation point is

$$R = \sqrt{r^2 + z^2}.$$

The field from that element scales like dq/R^2, but only the radial component survives after symmetry:

$$dE_r(z) \ \propto\ \frac{dq}{R^2}\cos\theta \quad \text{with} \quad \cos\theta = \frac{r}{R}.$$

Hence the radial contribution scales as

$$dE_r(z) \ \propto\ dq\,\frac{r}{(r^2+z^2)^{3/2}} = \lambda\,dz\,\frac{r}{(r^2+z^2)^{3/2}}.$$

Now observe the behaviour of the factor

$$f(z) = \frac{r}{(r^2+z^2)^{3/2}}:$$

- For $|z| \ll r$, $f(z) \approx \frac{r}{r^3} = \frac{1}{r^2}$ (elements near the closest point contribute like $1/r^2$).
- For $|z| \gg r$, $f(z) \approx \frac{r}{|z|^3}$, which decays very rapidly with $|z|$.

Interpretation: elements far away along the line ($|z| \gg r$) contribute negligibly to the *radial* field at the point. The dominant contribution comes from a "window" of length on the order of

$$|z| \lesssim \mathcal{O}(r).$$

So as you move farther away (increase r), the *effective contributing length* grows roughly $\propto r$. This is the crucial difference from a point charge (where there is no extra length to contribute).

3) Back-of-the-envelope scaling: why $1/r$ (not $1/r^2$)

Within the dominant window $|z| \lesssim r$, a typical element contributes on the order of

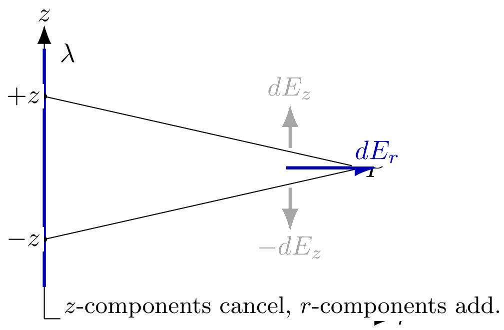
$$dE_r \sim \lambda\,dz \cdot \frac{1}{r^2}.$$

And the length of the window is $\Delta z \sim r$, so summing contributions over that window gives

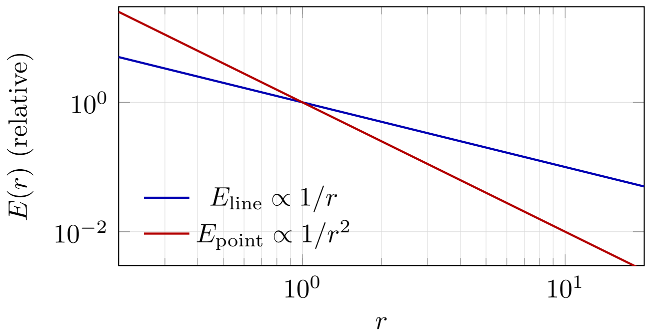
$$E(r) \sim \int_{-r}^{r} \lambda\,\frac{1}{r^2}\,dz \ \sim\ \lambda\,\frac{1}{r^2}\,(2r) \ \propto\ \frac{1}{r}.$$

Hence the field from a (very long) line charge decays more slowly:

$$\boxed{E_{\text{line}}(r) \propto \frac{1}{r}.}$$



Symmetry picture at P.



Decay rate: line charge is slower than point charge.

Figure 8: Intuition: far elements along the line contribute weakly in the radial direction, but the *effective contributing length* grows with r, producing $E \sim 1/r$.

4) Why a point charge decays faster $(1/r^2)$

A point charge has only one source point. There is *no* growing "effective contributing length" as r increases. So the only effect is geometric dilution of a single contribution:

$$\boxed{E_{\text{point}}(r) \propto \frac{1}{r^2}.}$$

Key Takeaways

- Pairing elements at $+z$ and $-z$ cancels the vertical components; radial components add.
- Far elements contribute weakly to the radial field, but the *effective contributing length* grows with distance r.
- Heuristic scaling: contribution per unit length behaves like $1/r^2$ over an effective length $\sim r$, giving $E \sim (1/r^2) \cdot r = 1/r$.
- For a point charge there is no growing contributing length, hence $E \propto 1/r^2$.